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MILITARY STANDARD

TEST REQUIREMENTS AND PROCEDURES FOR AIRCRAFT EMERGENCY GROUND AND DITCHING ESCAPE PROVISIONS

FSC 1506

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DEPARTMENT OF THE AIR FORCE
AIR FORCE SYSTEMS COMMAND
RESEARCH & TECHNOLOGY DIVISION
WRIGHT-PATTERSON AIR FORCE BASE, OHIO

Test Requirements and Procedures for Aircraft Emergency Ground and Ditching Escape Provisions

MIL-STD-872 (USAF)

1. This Military Standard is mandatory for use by the Department of the Air Force.

2. Recommended corrections, additions, or deletions should be addressed to the Systems Engineering Group (SEPSM), Wright-Patterson Air Force Base, Ohio 45433.

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CONTENTS

Paragraph		Page
1	SCOPE	1
1.1	Purpose	1
1.2	Application	1
2	REFERENCED DOCUMENTS	1
3	DEFINITIONS	1
3.1	Multiplace aircraft	1
3.2	Cargo aircraft	1
3.3	Cargo-transport aircraft	1
3.4	Simulated ground evacuation tests	1
3.5	Simulated ditching escape tests	1
4	GENERAL REQUIREMENTS	1
4.1	Test vehicle mockup	1
4.2	Test configurations	1
4.2.1	Multiplace and cargo aircraft	1
4.2.2	Cargo-transport aircraft	2
4.3	Test data	2
4.3.1	Documentary data	2
4.3.2	Photographic coverage	2
4.4	Test facilities	2
4.5	Test personnel	2
4.5.1	Personnel complement	2
4.5.1.1	Crew complement	2
4.5.1.2	Passenger complement	3
4.6	Test plan	3
4.7	Data requirements	3
5	DETAIL REQUIREMENTS	3
5.1	Test procedures	3
5.1.1	Walk-through inspection	3
5.1.2	Orientation	3
5.1.3	Identification of evacuees	3
5.1.4	Multiplace and cargo aircraft tests	3
5.1.4.1	Simulated ground evacuation (multiplace and cargo aircraft)	3
5.1.4.2	Simulated ditching escape (multiplace and cargo aircraft)	3

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CONTENTS (CONT'D)

Paragraph		Page
5.1.5	Cargo-transport aircraft tests	4
5.1.5.1	Simulated ground evacuation (cargo-transport aircraft)	4
5.1.5.2	Simulated ditching escape (cargo-transport aircraft)	4

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1. SCOPE

1.1 Purpose. This standard prescribes uniform testing requirements and procedures to be used for evaluating aircraft emergency ground and ditching escape provisions utilizing full-scale mockup.

1.2 Application. This standard is applicable to multiplace, cargo, and cargo-transport type aircraft. It covers the following types of tests:

a. Simulated ground evacuation tests

b. Simulated ditching escape tests.

2. REFERENCED DOCUMENTS

2.1 The issue of the following document in effect on the date of invitation for bids forms a part of this standard to the extent specified herein:

PUBLICATION

Air Force Systems Command Manual

AFSCM 80-1 Handbook of Instructions for Aircraft Design

3. DEFINITIONS

3.1 Multiplace aircraft. For the purpose of this standard, multiplace aircraft are classified as large combat- or trainer-type aircraft having three or more crewmembers.

3.2 Cargo aircraft. Cargo aircraft are those aircraft designed specifically to carry only cargo. Emergency ground and ditching escape provisions are required only for the crew on these aircraft.

3.3 Cargo-transport aircraft. Cargo-transport aircraft are those aircraft designed to carry cargo and passengers. Emergency ground and ditching escape provisions are required for both crew and passengers on these aircraft.

3.4 Simulated ground evacuation tests. Simulated ground evacuation tests are those tests conducted to demonstrate that sufficient numbers of doors, hatches, normal emergency exits, and emergency escape equipment are provided to permit complete personnel evacuation of the aircraft on the ground in accordance with the ground evacuation requirements specified in AFSCM 80-1.

3.5 Simulated ditching escape tests. Simulated ditching escape tests are those tests conducted to demonstrate that sufficient numbers of ditching escape hatches, ditching escape equipment, and adequate flotation and survival gear are provided to permit complete personnel evacuation of the aircraft in accordance with the ditching escape requirements specified in AFSCM 80-1 or by the procuring activity.

4. GENERAL REQUIREMENTS

4.1 Test vehicle mockup. The tests specified herein shall be conducted on a full-scale mockup of the test vehicle with proposed normal and emergency exits and entrances and proposed escape equipment and flotation gear installed and operational prior to final aircraft design.

4.2 Test configurations

4.2.1 Multiplace and cargo aircraft. Simulated ground evacuation tests shall be conducted using all exits. Simulated ditching escape tests shall be conducted using all ditching escape exits. Personnel representative of the crew of the test vehicle shall be used for these tests.

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Testing of cargo aircraft shall be accomplished with simulated cargo in the cargo compartment.

4.2.2 Cargo-transport aircraft. The following tests shall be accomplished from the configurations indicated using personnel representative of the crew and passengers of the test vehicle:

(a) Simulated ground evacuation tests

(1) Use all available exits (no cargo aboard, full passenger capacity).

(2) Use left-side exits (no cargo aboard, full passenger capacity.

(3) Use right-side exits (no cargo aboard, full passenger capacity).

(4) Use rear exits (no cargo aboard, full passenger capacity); use front exits under same conditions.

(5) Mixed configuration (repeat tests specified in 4.2.2(a) (2), (3), and (4) with cargo aboard; ratio and arrangement of passengers and cargo as defined by mission parameters.

(b) Simulated ditching escape tests

(1) Use all ditching escape exits (full passenger capacity, no cargo aboard).

(2) Mixed configuration (repeat test specified in 4.2.2(b) (1) with simulated cargo aboard; ratio and arrangement of passengers to cargo to be determined by the procuring activity).

4.3 Test data. Documentary and photographic data shall be recorded for all tests required by this standard. All documentary and photographic data shall be forwarded to the procuring activity.

4.3.1 Documentary data. Documentary data for each test shall include:

(a) Time required to open each emergency hatch or door

(b) Time required to activate, engage, or deploy the escape and survival equipment

(c) Number of personnel evacuating each 10 seconds for each exit, hatch, or door

(d) Total number of evacuees that use each exit

(e) Total time required to completely evacuate the aircraft.

4.3.2 Photographic coverage. Photoflash cameras, fixed and mobile movie cameras, and other photographic instrumentation as deemed necessary shall be used to obtain the photographic data. Film sizes and camera frame speed shall be appropriately selected to insure that clear and concise photographic data are provided.

4.4 Test facilities. The adequacy and safety of all equipment used in conducting the tests specified herein shall be approved by the procuring activity.

4.5 Test personnel

4.5.1 Personnel complement. Adequate personnel shall be provided to serve as crew, passengers, task manager, observers, equipment operators, and safety officers.

4.5.1.1 Crew complement. The crew complement shall consist of personnel representative of USAF flight crews. The physical stature of test personnel used as crewmembers shall be as specified by the procuring activity.

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4.5.1.2 Passenger complement. The passenger complement shall consist of personnel representative of passengers carried by USAF aircraft such as regular military passengers and combat (fully geared) and noncombat (lightly geared) troops. Representative passengers used in these tests should not be familiar with the aircraft configuration. If possible, passengers should not be reutilized in subsequent tests. The physical stature of test personnel used as passengers shall be as specified by the procuring activity.

4.6 Test plan. Before starting any tests under this standard a complete test plan outlining all proposed test procedures and the proposed test schedule for each aircraft shall be prepared. The test plan shall include test objectives, description of each test, equipment and facilities to be used, and data to be recorded. The test program shall not be started until written approval has been granted by the procuring activity.

4.7 Data requirements. The selected data requirements in support of this standard will be reflected in a Contractor Data Requirements List (DD Form 1423).

5. DETAIL REQUIREMENTS

5.1 Test procedures

5.1.1 Walk-through inspection. Prior to testing, a walk-through inspection of the test vehicle shall be conducted to observe if all exits are properly identified, accessible, and operable.

5.1.2 Orientation. Prior to each test, a briefing session shall be conducted with all test personnel to explain their particular mission and the test objectives.

5.1.3 Identification of evacuees. Test personnel used as evacuees shall each be provided with a distinct identification symbol to facilitate subsequent film analysis.

5.1.4 Multiplace and cargo aircraft tests

5.1.4.1 Simulated ground evacuation (multiplace and cargo aircraft). Personnel representative of the crew shall be assigned to their designated flight stations, indicating the particular exit, door, or hatch to be used during evacuation. Timers and photographic crew and equipment shall be appropriately positioned to cover all phases of the evacuation. The movie cameras shall be started prior to giving the signal to initiate evacuation. Evacuation shall be initiated. Documentary and photographic coverage shall be provided as specified herein. A minimum of three day and two night tests shall be conducted for each type of aircraft with the final number of tests and test conditions to be approved by the procuring activity.

5.1.4.2 Simulated ditching escape (multiplace and cargo aircraft). The same procedures shall be used as specified herein for the ground evacuation tests on multiplace and cargo aircraft except that the evacuees shall be assigned to their designated ditching stations, and evacuation shall be accomplished through the emergency ditching escape hatches. In addition, all flotation and survival gear shall be deployed. The deployment time shall be included in the total evacuation time. A minimum of two day and two night tests for each configuration specified in 4.2.2(b) (1) and (2) shall be conducted on each aircraft with the final number of tests and test conditions to be approved by the procuring activity.

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5.1.5 Cargo-transport aircraft tests

5.1.5.1 Simulated ground evacuation (cargo-transport aircraft). Personnel representative of the crew and passengers shall be assigned to their designated seating positions. The same procedures shall be used as specified herein for the ground evacuation tests on multiplace and cargo aircraft except as follows:

a. Conduct a minimum of two day tests for each configuration specified in 4.2.2(a) (1), (2), (3), and (4).

b. Conduct one day test for the configuration specified in 4.2.2(a) (5) and one night test for each configuration specified in 4.2.2(a) (2) and (5).

c. Conduct two additional tests (one day and one night test) for the configuration specified in 4.2.2(a) (2) but provide the test personnel with only a standard USAF preflight briefing (pointing out the location of the exits but not making individual exit assignments).

For the mixed configurations, the cargo and seating arrangements shall be altered and the evacuees rerouted to ascertain the most suitable escape pattern.

5.1.5.2 Simulated ditching escape (cargo-transport aircraft). The same procedures shall be used as specified herein for the ditching escape tests on multiplace and cargo aircraft except that a minimum of three day tests and one night test shall be conducted for each configuration specified in 4.2.2(b) (1) and (2). Two additional tests shall be conducted wherein the test personnel shall be provided with only a standard USAF preflight briefing (pointing out the location of the exits but not making individual exit assignments). The final number of tests and the test conditions shall be approved by the procuring activity.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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